

The Geometry of Polynomials in Algorithms

Lecture 6: Permanent and Van der Waerden

Lecturer: Nathan Klein

1 Capacity

Last time we studied the capacity of a polynomial $p \in \mathbb{R}[z_1, \dots, z_n]$ at a point $\beta \in \mathbb{R}_{\geq 0}^n$. Recall:

$$\text{cap}_{\beta}(p) = \inf_{z > 0} \frac{p(z)}{z^{\beta}}$$

We also said (but did not yet prove) that for a real stable polynomial p with non-negative coefficients, $\text{cap}_{\beta}(p) > 0$ for $\beta \in \mathbb{Z}_{\geq 0}^n$ if and only if the coefficient of the term z^{β} is non-zero. But the capacity also has meaning for general polynomials. In particular, the capacity at a point for a general polynomial with non-negative coefficients is non-zero exactly when that point is in the Newton polytope of p : the convex hull of the support of p .

Definition 1.1 (Newton Polytope). Given a polynomial $p \in \mathbb{R}_{\geq 0}[z_1, \dots, z_n]$, let $p(z_1, \dots, z_n) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} c_{\alpha} z^{\alpha}$. The Newton polytope of p is the convex hull of all α for which $c_{\alpha} > 0$. If $C_p \subseteq \mathbb{Z}_{\geq 0}^n$ is the set of α for which $c_{\alpha} > 0$, then:

$$\text{NewtonPolytope}(p) = \{\beta \in \mathbb{R}^n : \beta = \sum_{\alpha \in C_p} \lambda_{\alpha} \alpha \text{ for some } \lambda \in \mathbb{R}_{\geq 0}^{|C_p|}, \|\lambda\|_1 = 1\}$$

As discussed above, the capacity gives us a way to test for membership in the Newton polytope for any polynomial with non-negative coefficients, even when we don't have access to the coefficients explicitly:

Lemma 1.2. For any $p \in \mathbb{R}_{\geq 0}[z_1, \dots, z_n]$, $\beta \in \text{NewtonPolytope}(p)$ if and only if $\text{cap}_{\beta}(p) > 0$.

To prove this, we will need the weighted AM-GM inequality, which we recall now:

Lemma 1.3 (Weighted AM-GM). Let $\lambda \in \mathbb{R}_{\geq 0}^n$, $\|\lambda\|_1 > 0$. Then for any $x \in \mathbb{R}_{\geq 0}^n$:

$$\frac{\sum_{i=1}^n \lambda_i x_i}{\|\lambda\|_1} \geq \left(\prod_{i=1}^n x_i^{\lambda_i} \right)^{1/\|\lambda\|_1}$$

when $\|\lambda\|_1 = 1$, this has the simpler statement:

$$\sum_{i=1}^n \lambda_i x_i \geq \prod_{i=1}^n x_i^{\lambda_i}$$

Note that if $\lambda_i = \frac{1}{n}$ for all i , this recovers AM-GM.

Now let's prove [Lemma 1.2](#).

Proof. We will prove that if β is in the Newton polytope, then the capacity is positive, and leave the other direction as an exercise. Since β is in the polytope, we can write it as a convex combination of a set of vertices $S \subseteq \mathbb{Z}_{\geq 0}^n$ of the polytope.¹ So for some non-negative λ , $\|\lambda\|_1 = 1$:

$$\beta = \sum_{\alpha \in S} \lambda_{\alpha} \alpha$$

where WLOG $\lambda_{\alpha} > 0$ for all $\alpha \in S$. Now:

$$p(z) = \sum_{\alpha} c_{\alpha} z^{\alpha} \geq \sum_{\alpha \in S} c_{\alpha} z^{\alpha} = \sum_{\alpha \in S} \lambda_{\alpha} \frac{c_{\alpha} z^{\alpha}}{\lambda_{\alpha}} \geq \prod_{\alpha \in S} \left(\frac{c_{\alpha} z^{\alpha}}{\lambda_{\alpha}} \right)^{\lambda_{\alpha}} = z^{\beta} \prod_{\alpha \in S} \left(\frac{c_{\alpha}}{\lambda_{\alpha}} \right)^{\lambda_{\alpha}}$$

where in the last expression we used that $\beta = \sum_{\alpha \in S} \lambda_{\alpha} \alpha$. Therefore:

$$\text{cap}_{\beta}(p) = \inf_{z > 0} \frac{p(z)}{z^{\beta}} \geq \prod_{\alpha \in S} \left(\frac{c_{\alpha}}{\lambda_{\alpha}} \right)^{\lambda_{\alpha}} > 0$$

where here we use that all coefficients c_{α} are positive and $\lambda_{\alpha} > 0$ for all $\alpha \in S$. □

1.1 Computing the capacity

Last lecture, we said that the capacity together with Gurvits's theorem gives an e^n approximation for the permanent. However, we need to be able to compute the capacity (up to small error) in polynomial time. The main idea here is to do a change of variables and set $e^{z'_i} = z_i$, which we can do since we constrain $z > 0$ in our infimum. The problem is then (at the all 1s vector, since that is what's of interest at the moment):

$$\inf_{z'} \frac{p(e^{z'_1}, \dots, e^{z'_n})}{e^{\sum_{i=1}^n z'_i}}$$

We can also minimize the log instead, since log is a strictly increasing function. Then, we get:

$$\inf_{z'} \log p(e^{z'_1}, \dots, e^{z'_n}) - \sum_{i=1}^n z'_i$$

But $\log p(e^{z'_1}, \dots, e^{z'_n}) = \log(\sum_{\alpha} c_{\alpha} e^{\langle \alpha, z' \rangle})$. This is a log-sum-exp function, which is convex in z' , and subtracting an affine function maintains convexity. So we can use a convex program to find the capacity up to small multiplicative error.²

1.2 Gurvits' Theorem

Finally, let's show the theorem which lets us relate the capacity to the coefficient at β . We focus on the all 1s case here, but a similar theorem holds for general β .

Theorem 1.4 (Gurvits). *Let $p \in \mathbb{R}_{\geq 0}[z_1, \dots, z_n]$ be real stable. Then:*

$$e^{-n} \text{cap}_1(p) \leq \partial_{z_1} \dots \partial_{z_n} p|_{z=0} \leq \text{cap}_1(p)$$

¹By Carathéodory's theorem, we can choose S so that $|S| \leq n + 1$.

²Making this argument rigorous requires a few more details, which we omit here.

Proof. Let $c_1 = \partial_{z_1} \dots \partial_{z_n} p|_{z=0}$ be the coefficient of all 1s. First, let's prove that $c_1 \leq \text{cap}_1(p)$. This is simply because p has non-negative coefficients, so

$$\text{cap}_1(p) = \inf_{z>0} \frac{p(z)}{z_1 \dots z_n} \geq \inf_{z>0} \frac{c_1 z^1}{z^1} = c_1$$

Therefore this holds for all p with non-negative coefficients. The other side of the bound is not true of general polynomials, here we will crucially use stability. For example, even in the univariate case, consider $p(t) = 1 + t^2$. Then the coefficient of t is 0 but the capacity is $\inf_{t>0} \frac{1+t^2}{t} = \inf_{t>0} \frac{1}{t} + t \geq 2$ by AM-GM. However this polynomial is not real rooted.

The proof will proceed by induction, so first we'll prove it for the real rooted case.

Lemma 1.5. *Let $p \in \mathbb{R}_{\geq 0}[t]$ be real rooted. Then:*

$$p'(0) \geq \frac{1}{e} \inf_{t>0} \frac{p(t)}{t}$$

Proof. First, suppose $p(0) = 0$. But then:

$$\inf_{t>0} \frac{p(t)}{t} \leq \lim_{t \rightarrow 0} \frac{p(t)}{t} = p'(0)$$

so the lemma is true in this case.

Otherwise, let's write:

$$\log(p(t)) = \log(a_d \prod_{i=1}^d (t - \lambda_i)) = \log(a_d) + \sum_{i=1}^d \log(t - \lambda_i)$$

Since all λ_i are negative this is just the sum of $\log(t + c_i)$ for some $c_i \geq 0$. So this is the sum of concave functions, therefore it's concave on $[0, \infty)$. Now, taking the tangent at 0:

$$\log p(t) \leq \log p(0) + t(\log p)'(0) = \log p(0) + t \frac{p'(0)}{p(0)}$$

Therefore:

$$p(t) \leq e^{\log p(0) + t \frac{p'(0)}{p(0)}} = p(0) \exp\left(t \frac{p'(0)}{p(0)}\right)$$

If $p'(0) = 0$, then $p(t) \leq p(0)$ and:

$$\inf_{t>0} \frac{p(t)}{t} \leq \inf_{t>0} \frac{p(0)}{t} = 0$$

where the last equality is because we can take $t \rightarrow \infty$. So, we may assume $p'(0) > 0$ and we can take $t^* = \frac{p(0)}{p'(0)}$, which is non-zero and well defined. Then:

$$\inf_{t>0} \frac{p(t)}{t} \leq \frac{p(t^*)}{t^*} \leq \frac{p(0)}{t^*} e = p'(0)e$$

using our inequality from concavity, which completes the proof. $\square$

Now we can do the inductive step. Assume the statement holds for all polynomials on $n - 1$ variables. Now given $p(z_1, \dots, z_n)$, define the polynomial:

$$q(z_1, \dots, z_{n-1}) = \partial_{z_n} p \mid_{z_n=0}$$

q is real stable since partial differentiation and specialization are stability preservers.³ Now consider any fixed $a_1, \dots, a_{n-1} > 0$. Then the polynomial $p(a_1, \dots, a_{n-1}, t)$ is real rooted by specialization. Applying our lemma on the univariate case gives:

$$\partial_{z_n} p(a_1, a_2, \dots, a_{n-1}, 0) = q(a_1, \dots, a_{n-1}) \geq \frac{1}{e} \inf_{t>0} \frac{p(a_1, \dots, a_{n-1}, t)}{t}$$

therefore,

$$\begin{aligned} \text{cap}_1(q) &= \inf_{a_1, \dots, a_{n-1} > 0} \frac{q(a_1, \dots, a_{n-1})}{a_1 \dots a_{n-1}} \\ &\geq \inf_{a_1, \dots, a_{n-1} > 0} \left(\frac{1}{e} \inf_{t>0} \frac{p(a_1, \dots, a_{n-1}, t)}{a_1 \dots a_{n-1} t} \right) \\ &= \frac{1}{e} \text{cap}_1(p) \end{aligned}$$

Finally, by the induction hypothesis:

$$\partial_{z_1} \dots \partial_{z_n} p \mid_{z=0} = \partial_{z_1} \dots \partial_{z_{n-1}} q \mid_{z=0} \geq e^{-(n-1)} \text{cap}_1(q) \geq e^{-n} \text{cap}_1(p)$$

as desired. □

1.3 Van der Waerden

Finally, recall the Van der Waerden conjecture:

Theorem 1.6 (Van der Waerden Conjecture). *Let $M \in \mathbb{R}_{\geq 0}^{n \times n}$ be a doubly stochastic matrix, i.e., all row sums and column sums are 1. Then:*

$$\text{per}(M) \geq e^{-n}$$

in fact, one can obtain the slightly stronger $\text{per}(M) \geq \frac{n!}{n^n}$ with the slightly more precise proof of Gurvits' theorem, the original form of the conjecture.

To prove this, all we need to show is that for any doubly stochastic matrix, we have $\text{cap}_1(p) \geq 1$.

Lemma 1.7. *Let M be a doubly stochastic matrix, and let $p(z_1, \dots, z_n) = \prod_{i=1}^n (\sum_{j=1}^n M_{i,j} z_j)$. Then $\text{cap}_1(p) = 1$.*

Proof. We will use the weighted AM-GM again, noting that the $M_{i,j}$ sum to 1 across rows and columns.

$$p(z) = \prod_{i=1}^n \left(\sum_{j=1}^n M_{i,j} z_j \right) \geq \prod_{i=1}^n \prod_{j=1}^n z_j^{M_{i,j}} = z_1 \dots z_n$$

So the infimum is at least 1. We get equality because setting $p(\mathbf{1}) = 1$. □

³ q could also be zero, but then the capacity of p is 0 so there is nothing to prove, as I can take $z_n \rightarrow 0$.